

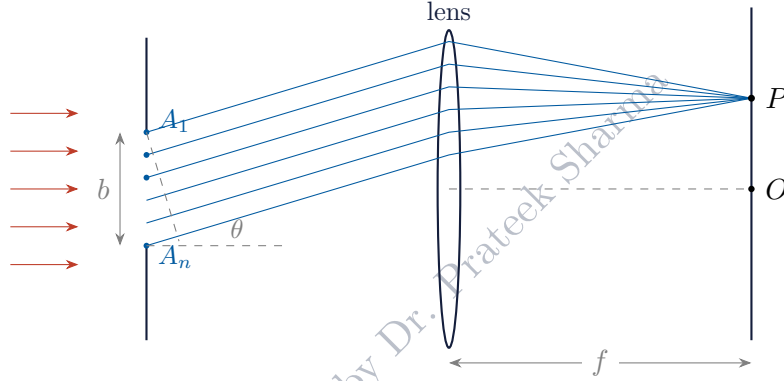
Fraunhofer Diffraction at a Single Slit — I

B.Sc. (V Semester) — Physical Optics · Kumaon University
Unit III: Diffraction · Lecture 2 · Instructor **Dr. Prateek Sharma**

Division of the slit into point sources · Sum of equally phased cosines · Resultant amplitude · Intensity distribution

1 Single-Slit Diffraction Pattern

A plane wavefront of monochromatic light is incident normally on a slit of width b . The light diffracted at an angle θ is focused by a lens of focal length f at a point P on the screen.



Divide the slit into n equally spaced point sources $A_1, A_2, \dots, A_n$, with spacing Δ , so that

$$b = (n - 1)\Delta.$$

The path difference between waves from two neighbouring points reaching P is

$$A_2A'_2 = \Delta \sin \theta,$$

and the corresponding phase difference is

$$\phi = \frac{2\pi}{\lambda} \Delta \sin \theta.$$

If each point source produces a field of amplitude a at P , the resultant field is

$$E = a [\cos \omega t + \cos(\omega t - \phi) + \cos(\omega t - 2\phi) + \dots + \cos(\omega t - (n - 1)\phi)].$$

2 Summing Equally Phased Cosines

We will show that

$$\sum_{k=0}^{n-1} \cos(\omega t - k\phi) = \frac{\sin(n\phi/2)}{\sin(\phi/2)} \cos\left[\omega t - \frac{1}{2}(n-1)\phi\right]$$

Write the sum as the real part of a sum of exponentials:

$$S = \sum_{k=0}^{n-1} \cos(\omega t - k\phi) = \operatorname{Re} \left\{ \sum_{k=0}^{n-1} e^{i(\omega t - k\phi)} \right\} = \operatorname{Re} \left\{ e^{i\omega t} \sum_{k=0}^{n-1} e^{-ik\phi} \right\}.$$

The remaining sum is a geometric progression (G.P.) of the form $a + ar + ar^2 + \dots + ar^{n-1}$:

$$\begin{aligned} S_n &= a + ar + ar^2 + \dots + ar^{n-1} \\ rS_n &= ar + ar^2 + ar^3 + \dots + ar^n \\ S_n - rS_n &= a - ar^n \\ \Rightarrow S_n &= \frac{a(1 - r^n)}{1 - r}. \end{aligned}$$

With $a = 1$ and $r = e^{-i\phi}$,

$$\sum_{k=0}^{n-1} e^{-ik\phi} = \frac{1 - e^{-in\phi}}{1 - e^{-i\phi}}.$$

Using the identity (for any angle x)

$$1 - e^{-ix} = e^{-ix/2} [e^{ix/2} - e^{-ix/2}] = e^{-ix/2} \cdot 2i \sin(x/2),$$

we get

$$\frac{1 - e^{-in\phi}}{1 - e^{-i\phi}} = \frac{e^{-in\phi/2} \sin(n\phi/2)}{e^{-i\phi/2} \sin(\phi/2)} = e^{-i(n-1)\phi/2} \frac{\sin(n\phi/2)}{\sin(\phi/2)}.$$

Therefore

$$S = \operatorname{Re} \left\{ \frac{\sin(n\phi/2)}{\sin(\phi/2)} e^{i[\omega t - \frac{(n-1)}{2}\phi]} \right\} = \frac{\sin(n\phi/2)}{\sin(\phi/2)} \cos \left[\omega t - \frac{(n-1)}{2}\phi \right].$$

3 Resultant Amplitude

The resultant field at P is therefore

$$\begin{aligned} E &= E_0 \cos \left[\omega t - \frac{(n-1)}{2}\phi \right], \\ E_0 &= a \frac{\sin(n\phi/2)}{\sin(\phi/2)}. \end{aligned}$$

Now let $n \rightarrow \infty$ and $\Delta \rightarrow 0$, keeping the slit width $b = (n-1)\Delta \approx n\Delta$ fixed. Then

$$\begin{aligned} \frac{n\phi}{2} &= \frac{\pi}{\lambda} n\Delta \sin \theta \rightarrow \frac{\pi}{\lambda} b \sin \theta, \\ \phi &= \frac{2\pi}{\lambda} \frac{b \sin \theta}{n} \rightarrow 0, \end{aligned}$$

so that $\sin(\phi/2) \approx \phi/2$, and

$$E_0 = \frac{a \sin(n\phi/2)}{\phi/2} = na \frac{\sin\left(\frac{\pi b \sin \theta}{\lambda}\right)}{\frac{\pi b \sin \theta}{\lambda}}.$$

$$E_0 = A \frac{\sin \beta}{\beta}, \quad A = na, \quad \beta = \frac{\pi b \sin \theta}{\lambda}$$

In the same limit $\frac{(n-1)}{2}\phi \rightarrow \beta$, so

$$E = A \frac{\sin \beta}{\beta} \cos(\omega t - \beta).$$

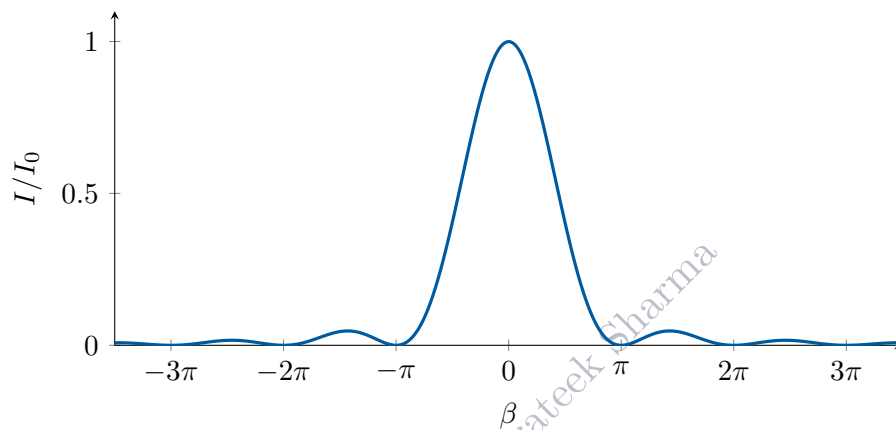
4 Intensity Distribution

The intensity is proportional to the square of the amplitude:

$$I = E_0^2 = A^2 \frac{\sin^2 \beta}{\beta^2}$$

$$I = I_0 \frac{\sin^2 \beta}{\beta^2}$$

where $I_0 = A^2$ represents the intensity at $\theta = 0^\circ$ (the centre of the pattern).



(Single-slit intensity distribution. The central maximum is twice as wide as the secondary maxima, which are much weaker. Their positions are found in Lecture 3.)